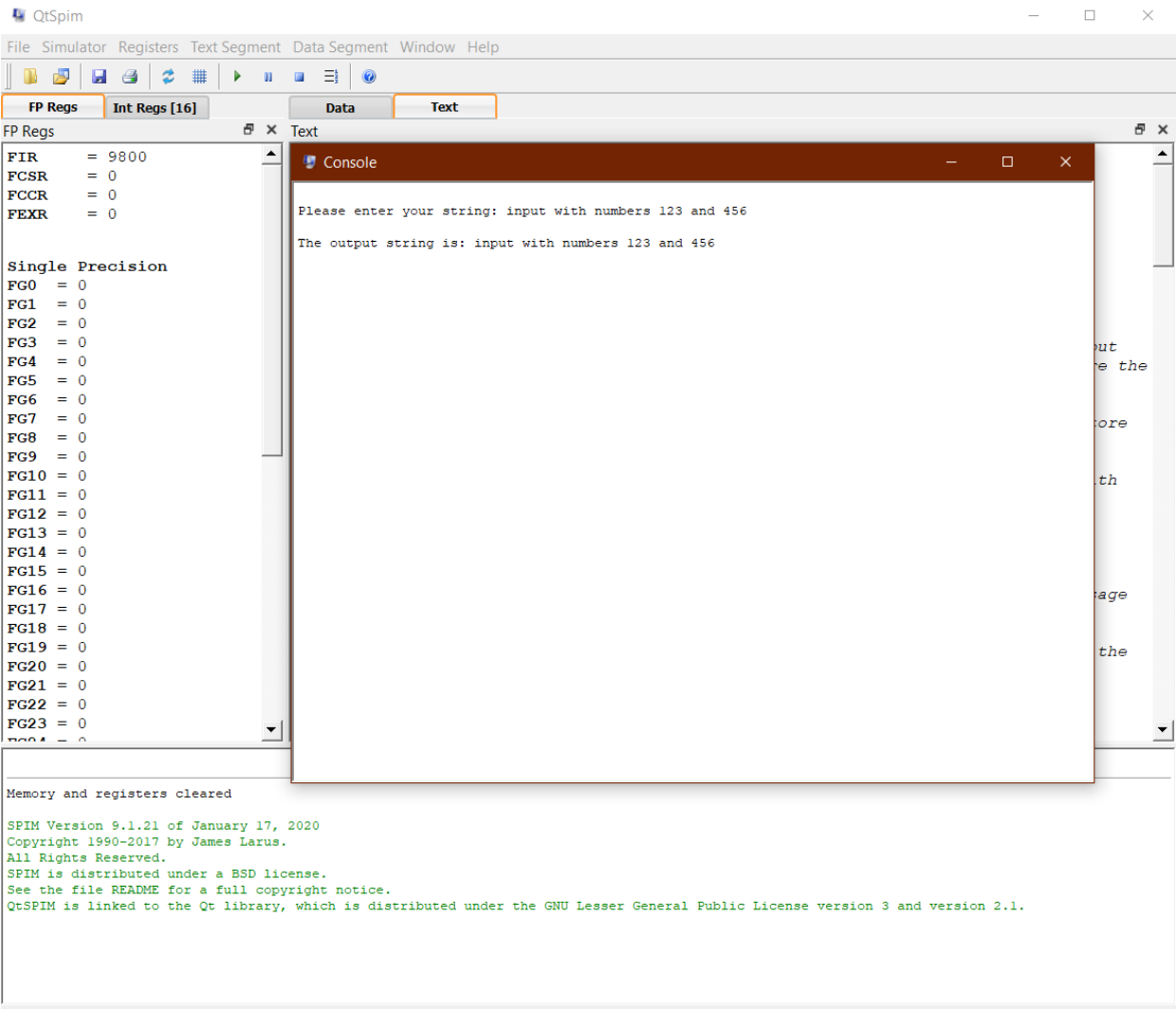


Το πρόγραμμα που έχουμε ως αρχείο .asm αφήνει τα πεζά και τους αριθμούς ως έχουν, μετατρέπει τα κεφαλαία σε πεζά και επίσης μετατρέπει τα σύμβολα σε κενούς χαρακτήρες. Τα αποτελέσματα της κονσόλας φαίνονται παρακάτω.



The screenshot shows the QtSPIM simulator window. The 'Text' tab is selected, and the 'Console' window is open. The console displays the following text:

```
Please enter your string: input with numbers 123 and 456
The output string is: input with numbers 123 and 456
```

The main window also shows the 'FP Regs' and 'Int Regs [16]' tabs. The 'FP Regs' tab is active, displaying the following values:

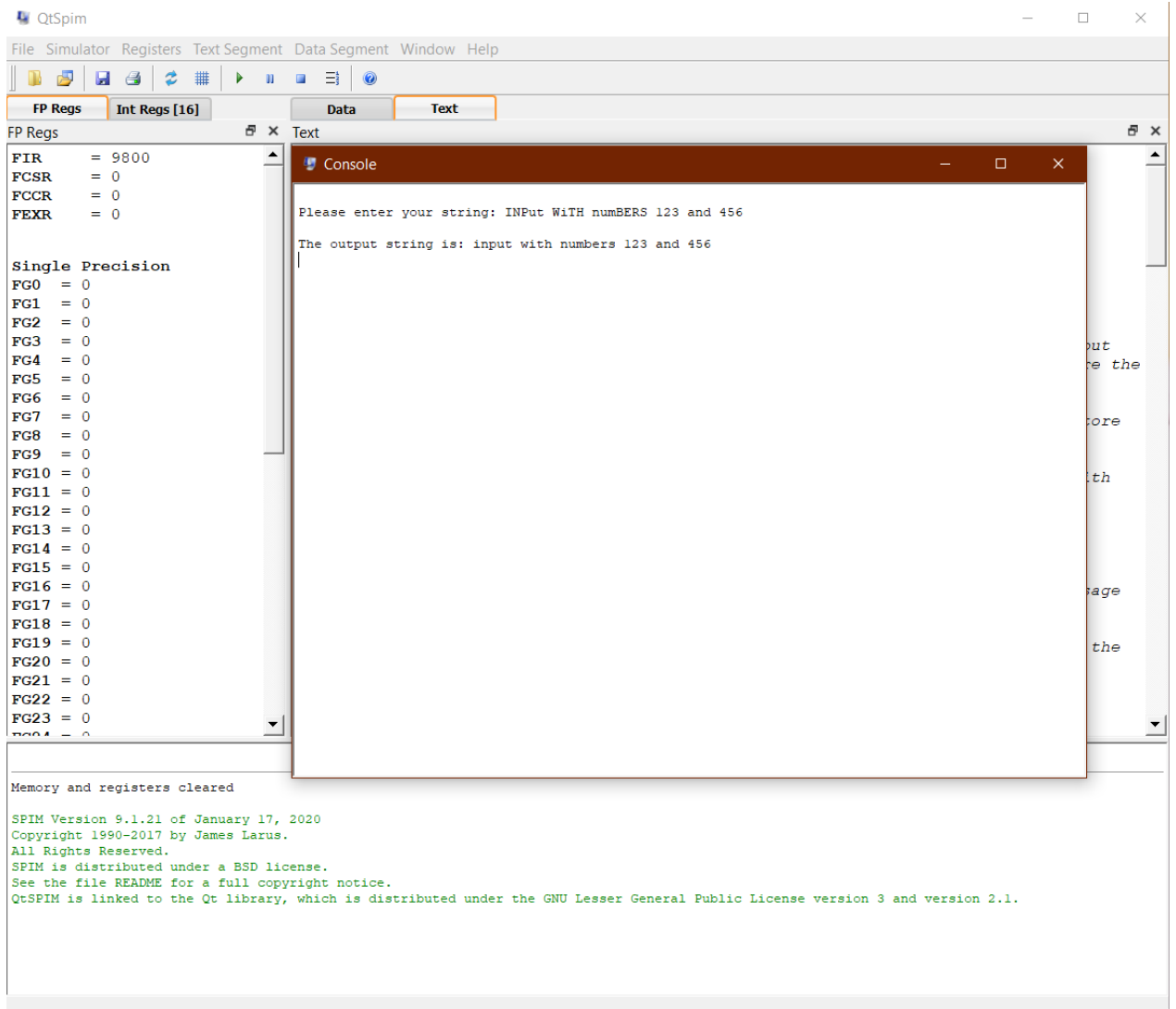
```
FIR = 9800
FCSR = 0
FCCR = 0
FEXR = 0
```

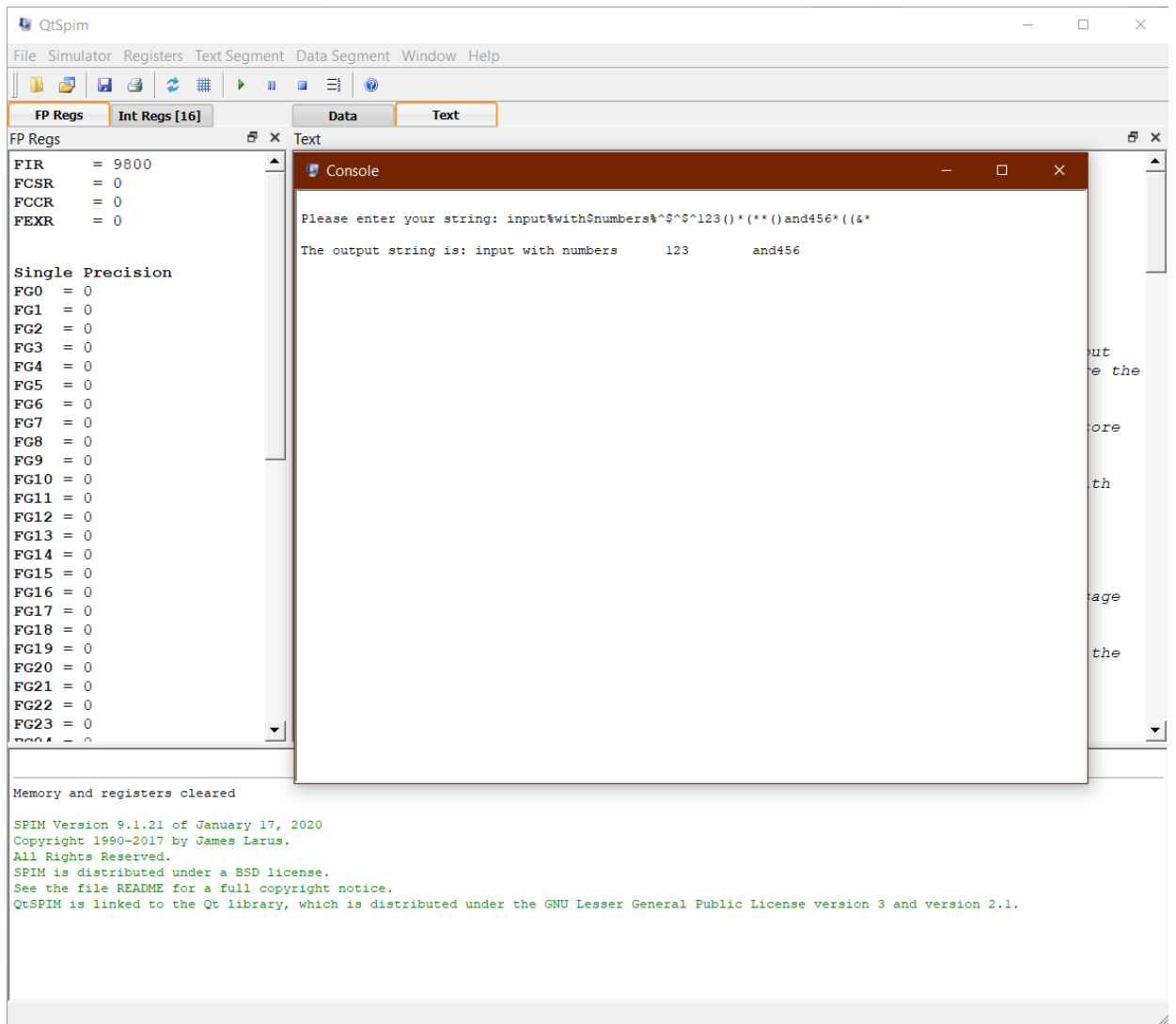
The 'Single Precision' section shows the following values:

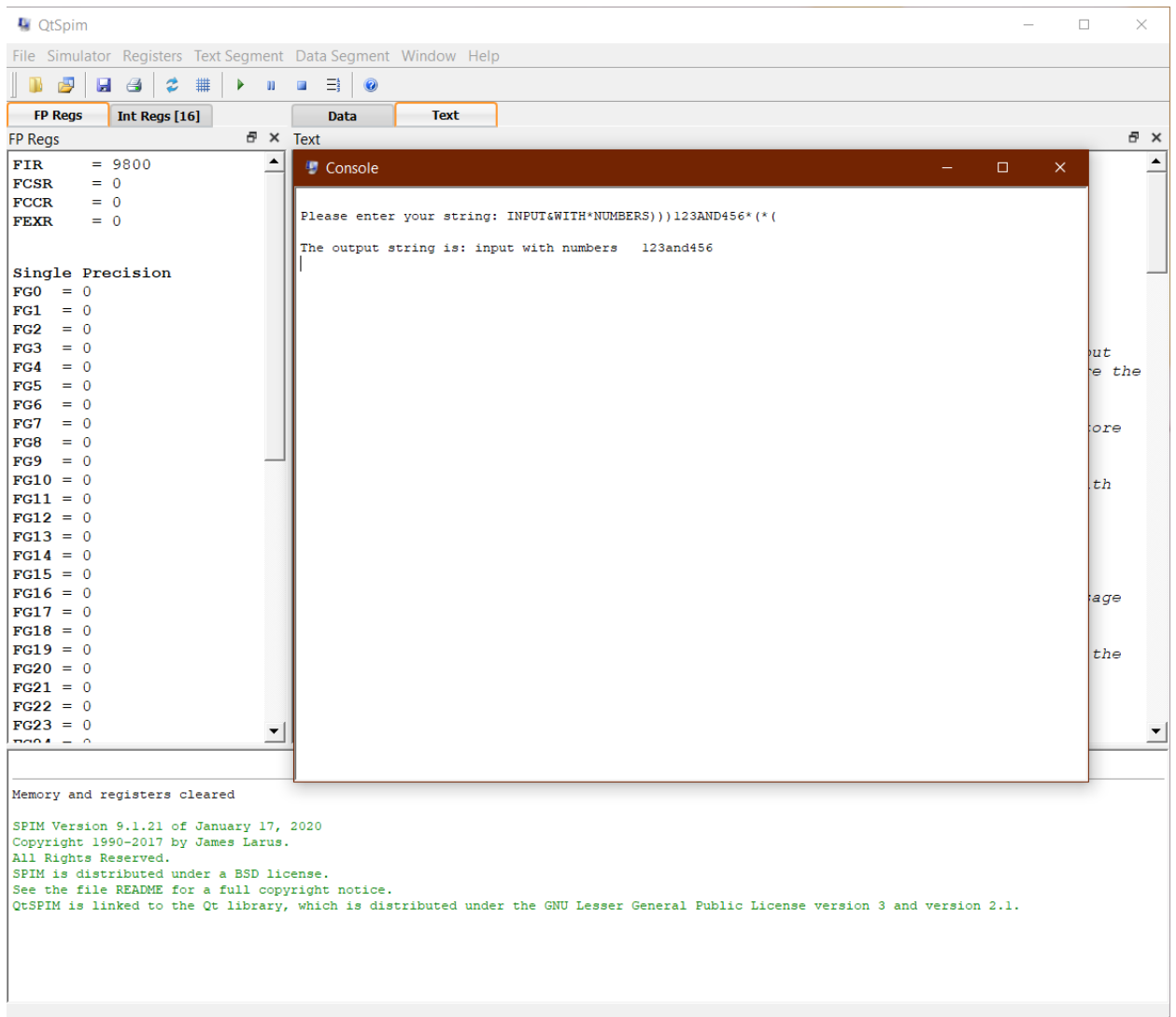
```
FG0 = 0
FG1 = 0
FG2 = 0
FG3 = 0
FG4 = 0
FG5 = 0
FG6 = 0
FG7 = 0
FG8 = 0
FG9 = 0
FG10 = 0
FG11 = 0
FG12 = 0
FG13 = 0
FG14 = 0
FG15 = 0
FG16 = 0
FG17 = 0
FG18 = 0
FG19 = 0
FG20 = 0
FG21 = 0
FG22 = 0
FG23 = 0
```

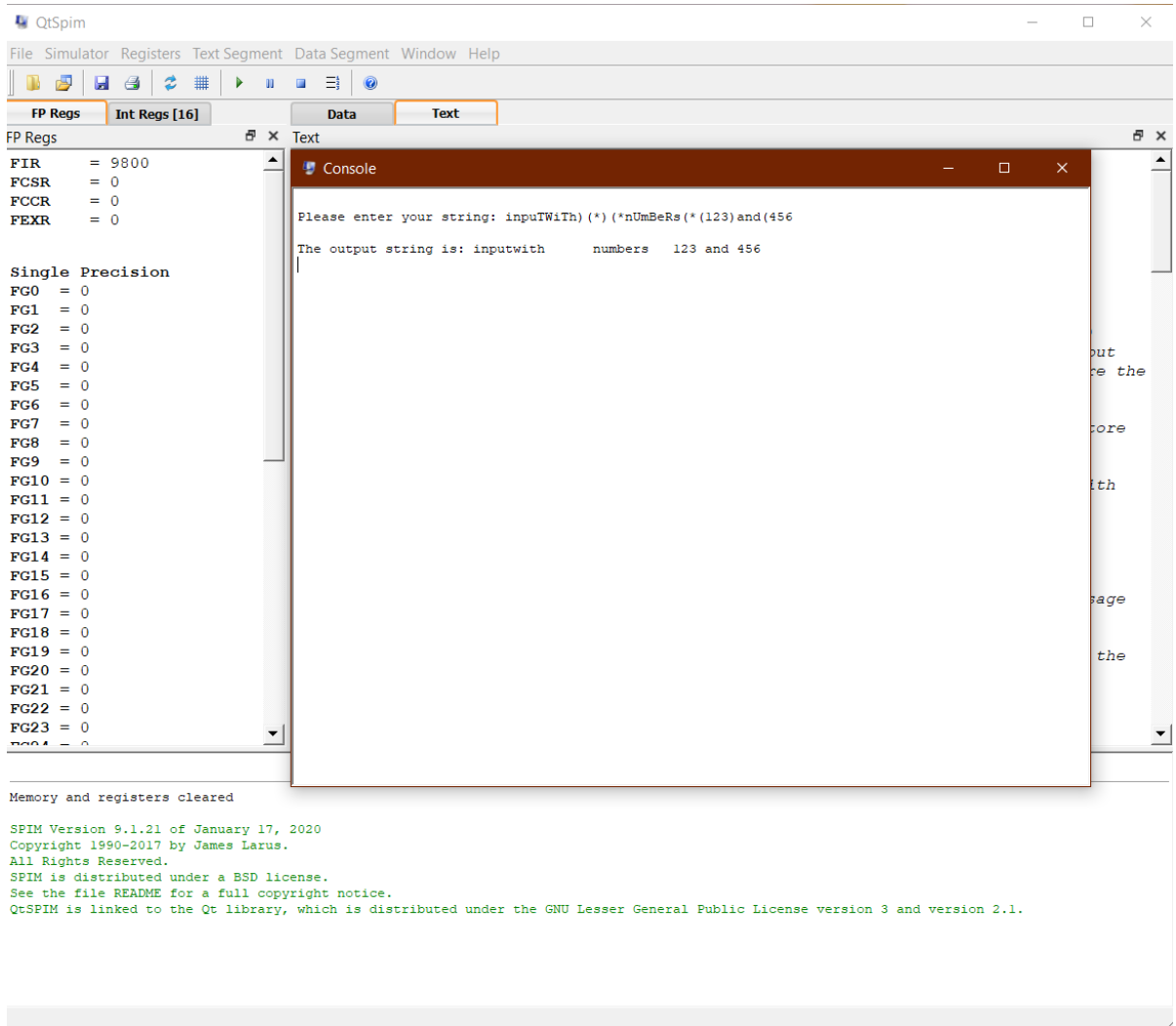
The bottom status bar displays the following text:

```
Memory and registers cleared
SPIM Version 9.1.21 of January 17, 2020
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QtSPIM is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
```









Ωστόσο, έχουμε ένα αρχείο .txt με κώδικα, ο οποίος σκοπό έχει να κάνει τα παραπάνω και επιπλέον να διαγράφει τους έξτρα κενούς χαρακτήρες. Ο κώδικας αυτός δεν τρέχει σωστά, όπως φαίνεται και στο αποτέλεσμα της κονσόλας.

QtSpim

File Simulator Registers Text Segment Data Segment Window Help

Int Regs [16] FP Regs

Int Regs [16]

PC = 40004c
EPC = 0
Cause = 0
BadVAddr = 0
Status = 3000fff10
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 10010000
R2 [v0] = a
R3 [v1] = 0
R4 [a0] = 100100fc
R5 [a1] = 10010098
R6 [a2] = 100100fc
R7 [a3] = 0
R8 [t0] = 10010034
R9 [t1] = a
R10 [t2] = 0
R11 [t3] = a
R12 [t4] = 10010098
R13 [t5] = 0
R14 [t6] = 100100fc
R15 [t7] = 0
R16 [s0] = 0
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0

Unaligned address in inst/data fetch: 0x100100a3

Memory and registers cleared

SPIM Version 9.1.21 of January 17, 2020
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SPIM is distributed under a BSD license.
See the file README for a full copyright notice.
QtSpim is linked to the Qt library, which is distributed under the GNU Lesser General Public License version 3 and version 2.1.
Exception occurred at PC=0x00400118
Unaligned address in inst/data fetch: 0x10010098

Memory and registers cleared

User Text Segment [00400000]..[00440000]

[00400000] 8fa40000 lw \$4, 0(\$29) ; 183: lw \$a0 0(\$sp) # argc
[00400004] 27a50004 addiu \$5, \$29, 4 ; 184: addiu \$a1 \$sp 4 # argv
[00400008] 24a60004 addiu \$6, \$5, 4 ; 185: addiu \$a2 \$a1 4 # envp
[0040000c] 00041080 sll \$2, \$4, 2 ; 186: sll \$v0 \$a0 2
[00400010] 00c23021 addu \$6, \$6, \$2 ; 187: addu \$a2 \$a2 \$v0
[00400014]
[00400018]
[0040001c]
[00400020]
[00400024]
[00400028]
[0040002c]
[00400030]
[00400034]
[00400038]
[0040003c]
[00400040]
[00400044]
[00400048]
[0040004c]
[00400050]
[00400054]
[00400058]
[0040005c]
[00400060]
[00400064]
[00400068]
[0040006c]
[00400070]
[00400074]

Please enter your string: input W!Th (*)(*nUmBeRs (* (123)and(456
The output string is:

ss of the input_string to \$a0
address of the process_string to \$a1
ress of the output_string to \$a2
ts \$a0, \$a1 and \$a2
t_string to \$t0 (to process the string)

11:11 μμ
18/11/2020